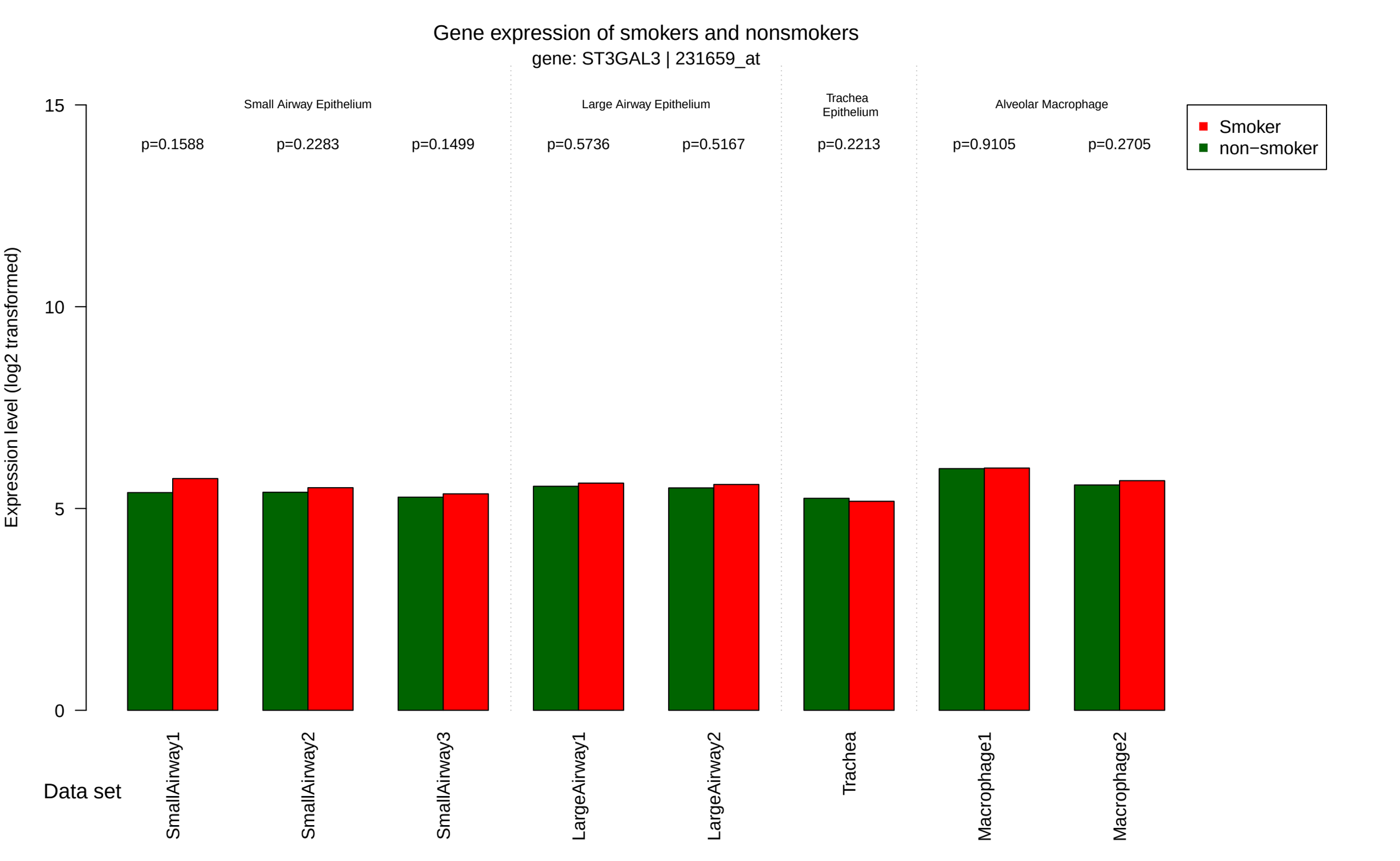
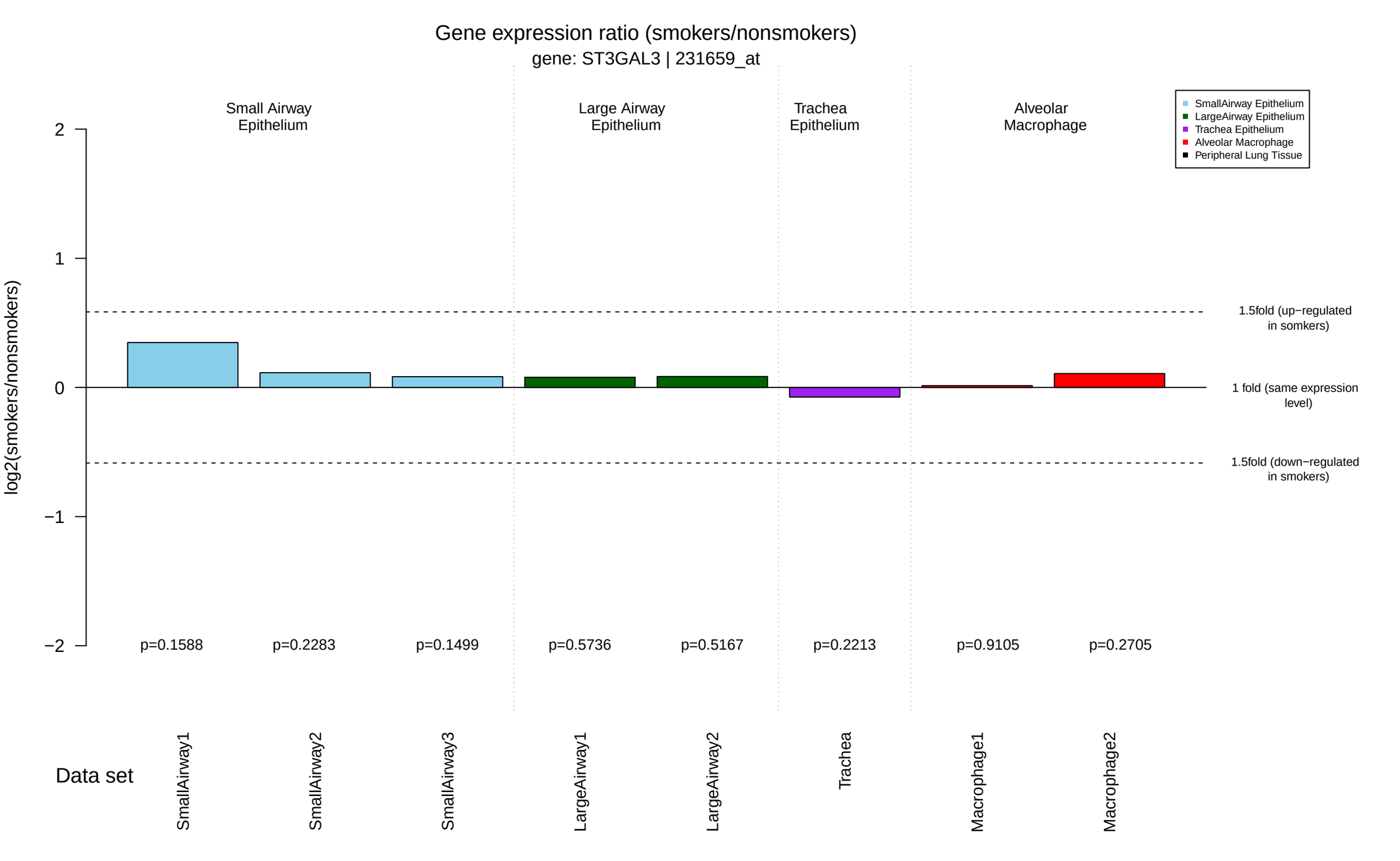
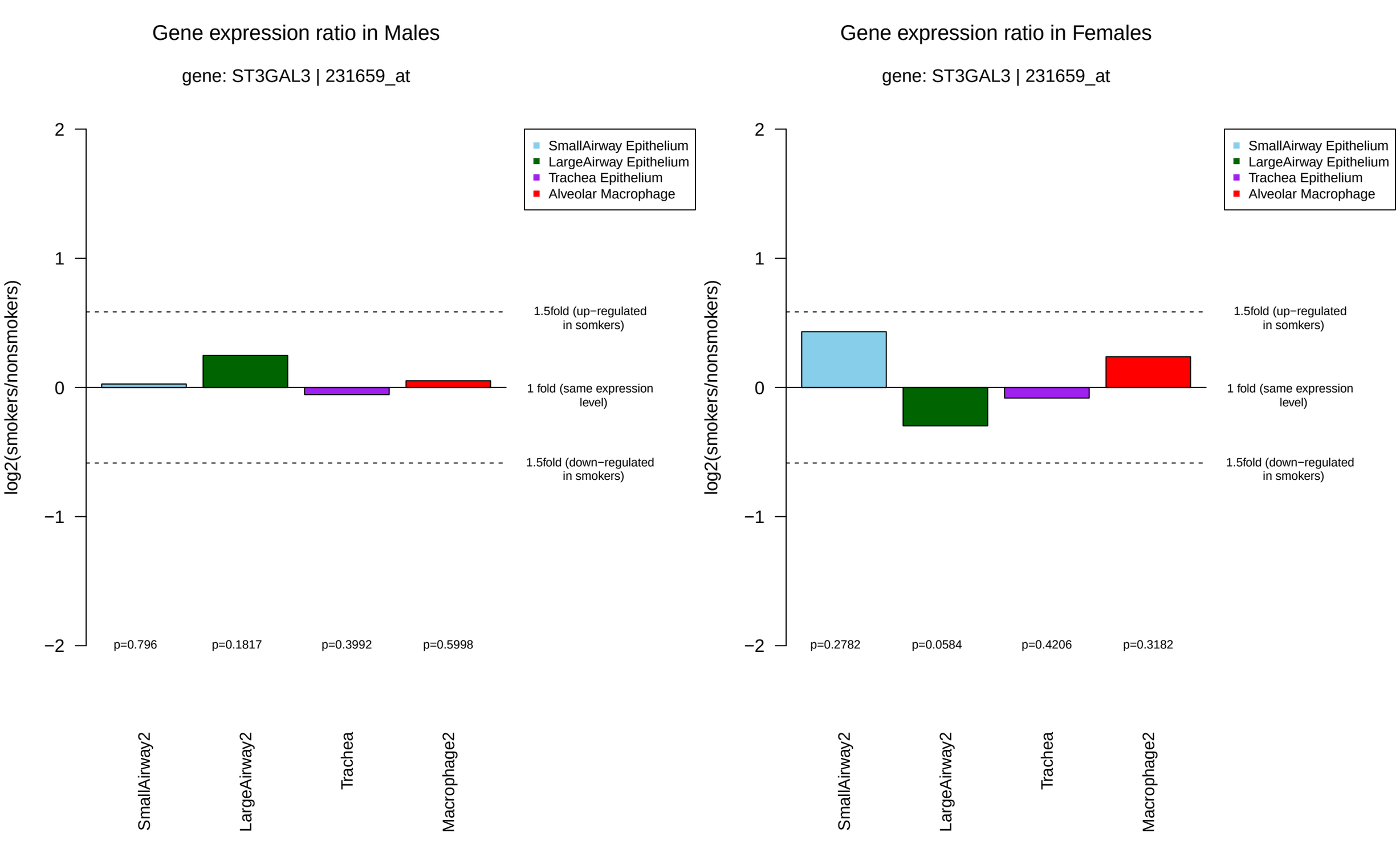
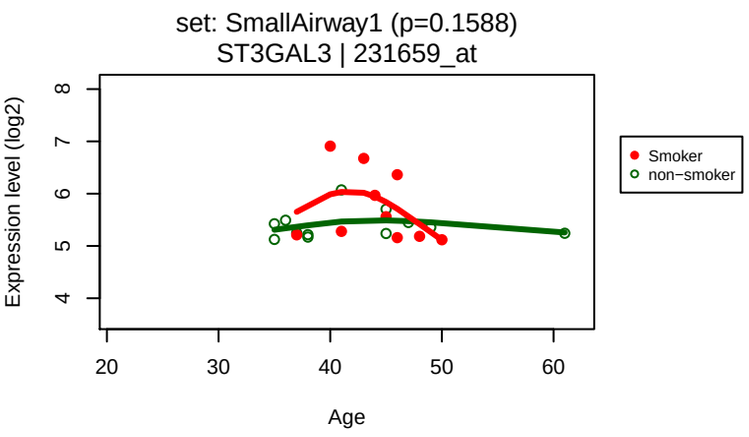
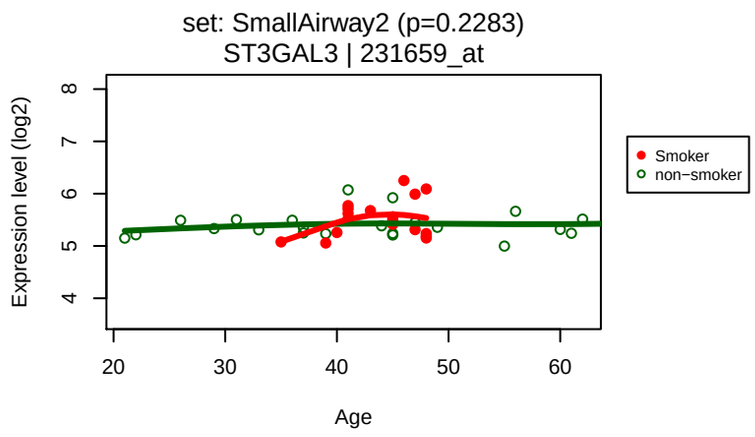
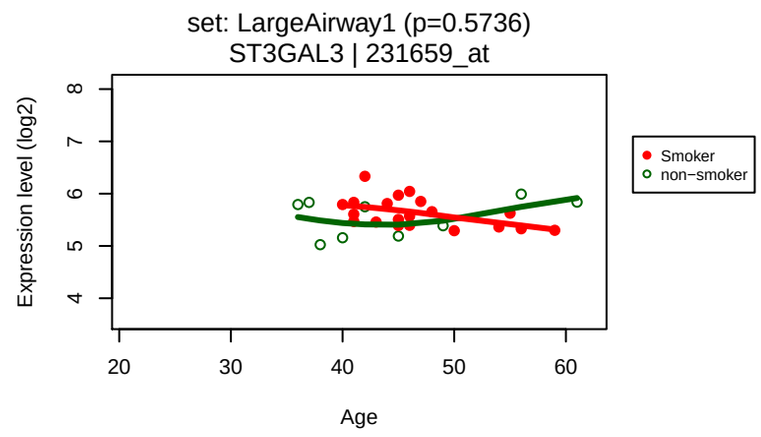

Gene expression of smokers and nonsmokers
gene: ST3GAL3 | 231659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1588
p=0.2283
p=0.1499
p=0.5736
p=0.5167
p=0.2213
p=0.9105
p=0.2705
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

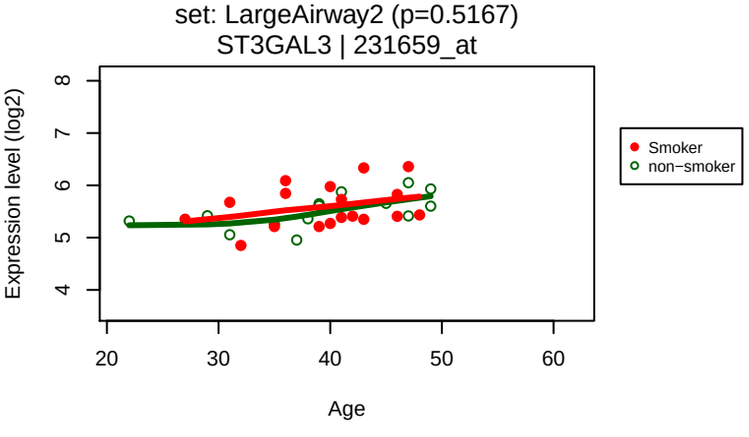

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL3 | 231659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1588
p=0.2283
p=0.1499
p=0.5736
p=0.5167
p=0.2213
p=0.9105
p=0.2705
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

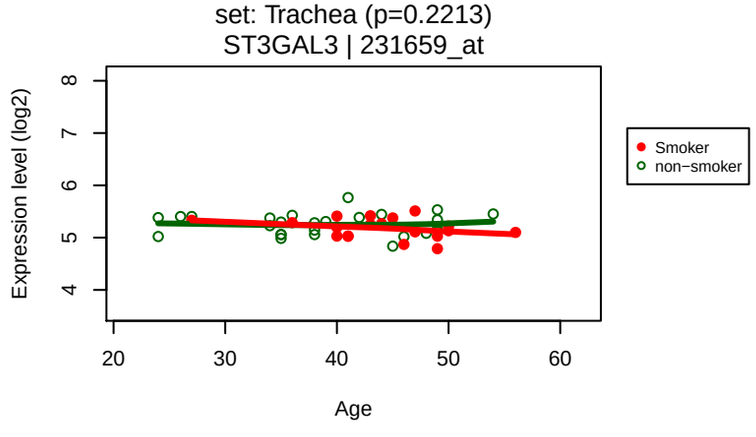

Gene expression ratio in Males
gene: ST3GAL3 | 231659_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.796
p=0.1817
p=0.3992
p=0.5998
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL3 | 231659_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2782
p=0.0584
p=0.4206
p=0.3182
SmallAirway2
LargeAirway2
Trachea
Macrophage2

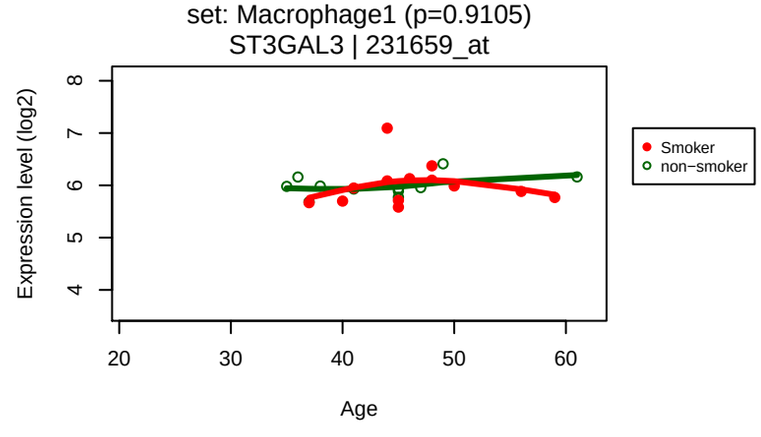

set: SmallAirway1 (p=0.1588)
ST3GAL3 | 231659_at
Expression level (log2)
Age
Smoker
non−smoker

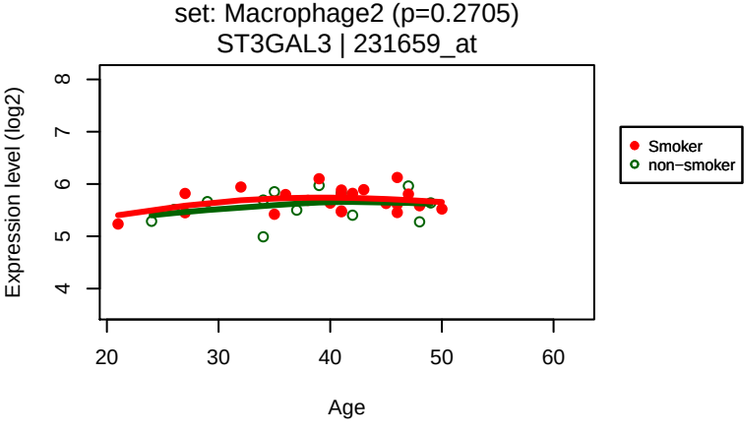

set: SmallAirway2 (p=0.2283)
ST3GAL3 | 231659_at
Expression level (log2)
Age
Smoker
non−smoker

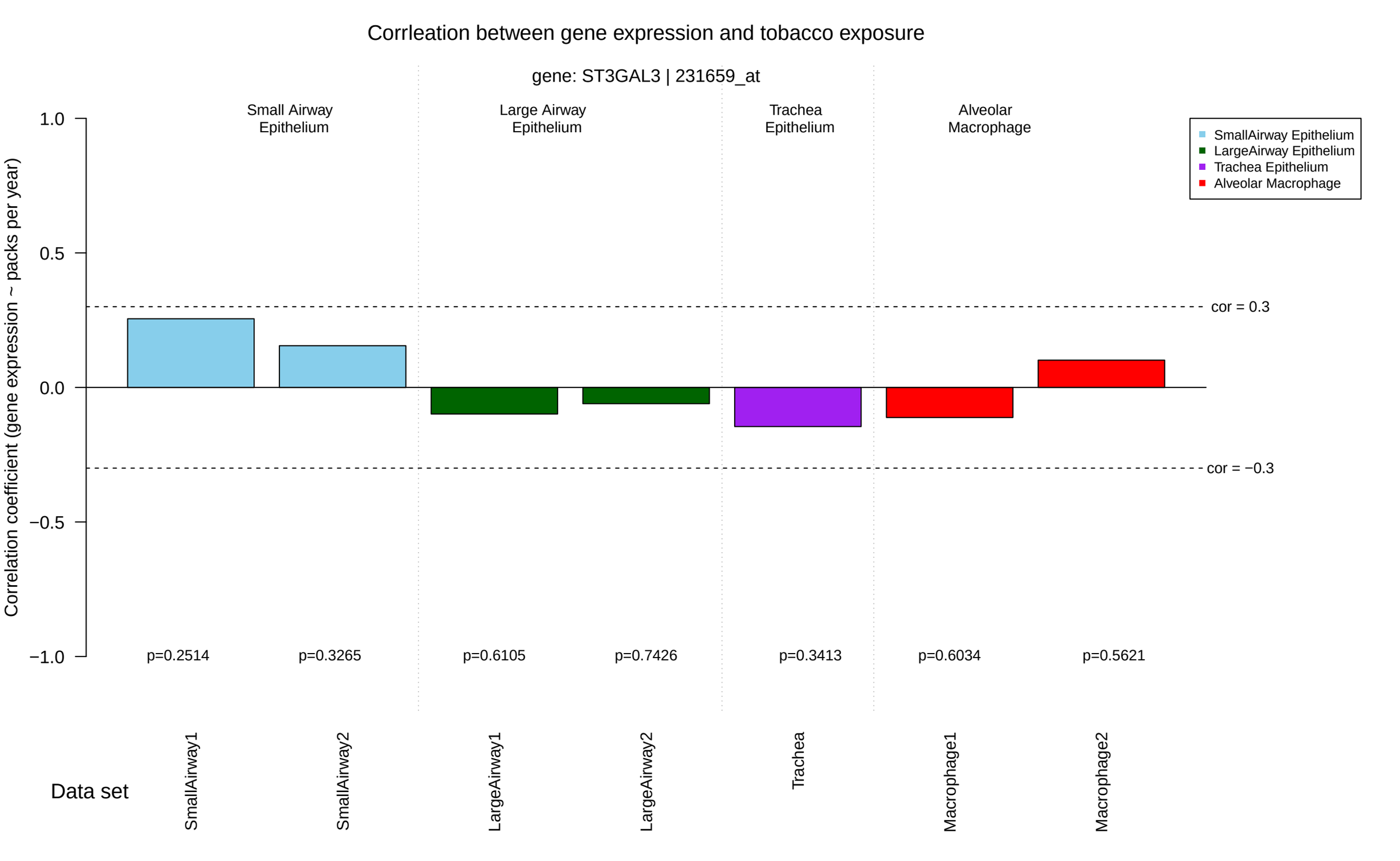

set: LargeAirway1 (p=0.5736)
ST3GAL3 | 231659_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5167)
ST3GAL3 | 231659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2213)
ST3GAL3 | 231659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9105)
ST3GAL3 | 231659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2705)
ST3GAL3 | 231659_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL3 | 231659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2514
p=0.3265
p=0.6105
p=0.7426
p=0.3413
p=0.6034
p=0.5621
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2